MATH 128a: Programming assignment #2

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Due by: **Thursday, October 23, 2025, 11:59pm**

Problem 1: Damped Newton iteration

In this problem, we consider a modification of Newton's method to increase its radius of convergence.

(a) Download the Matlab function `newton_1d.m` from bcourses. Modify it to output not just the final approximation of the root of f but also the intermediate approximations p_i $i = 0, 1, 2, \dots$ that are computed as well as the corresponding residuals (errors). Then run it with the input $f(x) = \tan^{-1}(x)$, tolerance $\epsilon = 10^{-6}$ and

(i) $p_0 = 1.35$

(ii) $p_0 = 1.45$.

For each of them, create a plot that displays f and the first 4 approximations computed.

(b) Implement the damped Newton method

$$p_{n+1} = p_n - \lambda_n \frac{f(p_n)}{f'(p_n)}, \quad n = 0, 1, 2, \dots,$$

where λ_n is chosen from $\{1, \frac{1}{2}, \frac{1}{4}, \dots, \lambda_{min}\}$ using the following monotonicity test: At each $n = 1, 2, \dots$, we start with $\lambda_n = \min\{1, 2\lambda_{n-1}\}$ and check whether

$$|(f'(p_n))^{-1} f(p_{n+1})| \leq (1 - \lambda_n/2)|(f'(p_n))^{-1} f(p_n)|. \tag{1}$$

If yes, we take this λ_n, if no, we try the test (1) with $\lambda_n/2$, if it is satisfied, we take this one, else we again decrease it by a factor $1/2$, etc. If $\lambda_n < \lambda_{min}$, we stop the iteration and restart with a new p_0 (print an error). In the first iteration, we start testing (1) with $\lambda_0 = 1$. Write a function that returns the intermediate approximations, their residuals and the step sizes used.

Then test the method for

(i) $f(x) = \tan^{-1}(x)$, $x \in \mathbb{R}$, starting values $p_0 = 2$ and $p_0 = 20$.

(ii) $f(x) = x^k$, $x \in \mathbb{R}$ and $k = 4, 8, 16$, $p_0 = 2$, $p_0 = 50$.

(iii) $f(x) = x^{1/(2k+1)}$, $k = 2, 4, 8$, $p_0 = 5$, $p_0 = 20$.

Set $\lambda_{min} = 0.001$. How many iterations do you need in each case to achieve an error of 10^{-6} and how does it compare to the standard Newton method?

Problem 2: Fixed point, secand and Newton method

A ball with radius $R = 1$ and density $\rho \in (0, 1)$ is floating on the water (density of the water is $\rho_w = 1$). Find the depth $h = h(\rho)$ of this ball (measured from the lowest point sunk into the water).

(a) Show using the balance of forces that $h^3 - 3h^2 + 4\rho = 0$ holds.

(b) Solve this equation using the fixed point iteration $h_{k+1} = \sqrt{(h_k^3 + 4\rho)/3}$, using the secant method and Newton's method for $\rho = 0.0001, 0.05, 0.4, 0.6, 0.95, 0.9999$. Use $h_0 = R$ as a starting value. Terminate the iterations as soon as $|h_k - h_{k-1}| \leq 10^{-5}$.

(c) Explain the observed orders of convergence in dependence of ρ.

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